

Hansika Karunathilake

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SUMMARY

Software Engineer with nearly 1.5+ years of experience bridging full-stack development expertise with a passion for software quality. Skilled in building scalable web applications and shipping end-to-end automated test suites for production-facing systems across web, mobile, API, and performance layers. Backed by a strong Java/OOP foundation and proficiency in modern JS/TS frameworks, I have proven experience integrating robust test automation directly into CI/CD pipelines. Commended for exemplary initiative and teamwork, I bring a developer's mindset to problem-solving and bug isolation, aiming to deliver reliable, high-performing applications.

EDUCATION

University of Ruhuna, Faculty of Engineering | Sri Lanka

B.Sc. (Hons) in Computer Engineering | Expected Sept 2026 | Current GPA: 3.8/4.0

EXPERIENCE

Software Engineering Intern | WSO2 | Sri Lanka | May 2025 - Nov 2025

Global technology company specializing in enterprise integration, API management, and developer-focused solutions.

- Engineered the official Ballerina Connector for Amazon SQS, implementing core APIs and an event-driven listener for asynchronous message polling.
- Delivered 18 production APIs with 96% automated test coverage, writing 58 comprehensive test cases and significantly reducing manual QA effort.
- Designed and built an end-to-end AI-powered automation pipeline utilizing Anthropic LLMs to generate Ballerina connectors from OpenAPI specifications.
- Automated OpenAPI sanitization (schema renaming, operationId generation), AI-powered test generation, document generation, and example generation successfully presenting the full workflow to the Integration team and stakeholders.
- Tools/Technologies:** Ballerina, Java, AWS SQS, Anthropic LLM, Python, OpenAPI, GitHub Actions

Freelance Full-Stack Developer | Self-Employed | Sri Lanka | July 2024 – Present

- Developed scalable web applications for multiple clients, including a boutique restaurant, travel & tour platforms.
- Designed and maintained automated regression suites using Playwright for booking and content flows, successfully catching regressions pre-deployment across three client platforms.
- Collaborated across full project lifecycles, from architecture to release, translating business needs into secure, high-performing software.
- Tools/Technologies:** NextJS, NestJS, React, PostgreSQL, Prisma, Playwright, Tailwind CSS, AWS S3

Founding Committee Member (Secretary) & Full Stack Developer | ICORE – University of Ruhuna | Sri Lanka | April 2025 – May 2026

Official university-approved innovation club and digital platform enabling engineering undergraduates to publish portfolios and technical case studies.

- Collaborated within an 8-member software engineering team to architect and build a scalable digital portfolio ecosystem from the ground up.
- Contributed to frontend and backend development within Agile workflows, utilizing Jira for task management and adhering to strict Git version control standards.
- Acted as Club Secretary for the founding committee, managing human resources, team coordination, and administrative operations.
- Supported the successful onboarding and showcasing of 40+ undergraduate engineering projects on the platform.
- Tools/Technologies:** JavaScript/TypeScript, Git, Jira, Agile/Scrum, Next.js, NestJS

PROJECTS DELIVERED

Blockchain-Based, Privacy-Preserving Reporting Service for Local Governance (PoC) – Final Year Project ([Link](#))

- **Smart Contract development:** Authored and deployed the Solidity-based OpinionPolling smart contract. Orchestrated contract deployment on a private, permissioned Proof-of-Authority (PoA) Geth network.
- **AI Content Moderation:** Developed a containerized AI oracle network (Safety, Spam, Civic Relevance) utilizing pre-trained NLP and computer vision models (BERT, Falconsai) to filter reports before immutable on-chain storage.
- **End-to-End Quality Engineering:** Formulated and executed the master QA test plan. Authored unit tests (Hardhat/Chai for Solidity, Jest for NestJS backend) and built a custom Python test client to validate security, boundary, and classification test cases.
- **Adversarial & Security Testing:** Validated critical security boundaries including replay attack mitigation (via ZKP ticket nullifier tracking), relayer signature authentication (ECDSA), and cryptographic payload validation.
- **Performance Benchmarking:** Stress-tested the private PoA Clique blockchain consensus, measuring transactions per second, block finality time, and IPFS upload/retrieval latency.
- **Research Publication:** Authored and submitted a research paper as the first author to the 2026 IEEE International Conference on Electrical Engineering and Emerging Technologies (ICEEET) (Currently under review).
- **Tools/Technologies:** Solidity, Go-Ethereum (Geth), Hardhat, Chai, Jest, Playwright, Python, Pytest, Docker Compose, Next.js, NestJS, IPFS, VPS/Linux

CeylonTrips – Multilingual Travel & Booking Platform ([Link](#))

- **Platform Architecture & Development:** Collaborated within a 3-member team to gather client requirements and architect a dynamic tour platform. Developed dynamic round-tour pages, several optimized static pages, and an interactive form to capture tailor-made tour requests from travelers. Built a secure admin dashboard for seamless tour management, and integrated a custom blog utilizing Sanity CMS hosted and deployed on a VPS.
- **Internationalization & SEO:** Engineered language translation across major pages into 10 languages using i18n internationalisation, successfully driving organic traffic with over 200 pages currently indexed on Google.
- **Testing & Quality:** Designed and integrated automated UI testing suites using Playwright and the Page Object Model into CI pipelines, ensuring reliability across tour and destination flows.
- **Impact:** Delivered a highly scalable and SEO-optimized travel platform that successfully expanded the client's global reach through multilingual support. Ensured a flawless user experience and prevented live defects by consistently catching UI regressions pre-deployment.
- **Tools/Technologies:** Next.js, NestJS, Neon PostgreSQL, Prisma ORM, Sanity CMS, Playwright, i18n, VPS.

SKILLS

Programming: Java, JavaScript, TypeScript, Python, Ballerina, C++, Solidity

Web Frameworks & Libraries: Next.js, NestJS, React.js, Node.js, Express.js, GraphQL, Tailwind CSS, Flutter, i18n.

Testing & Quality Engineering: Playwright (POM), Selenium WebDriver, Appium, REST Assured, k6, JMeter, JUnit, Postman.

Databases & ORMs: PostgreSQL, MongoDB, Prisma ORM

DevOps & Infrastructure: Docker, AWS (SQS, S3), CI/CD (Jenkins, GitHub Actions), VPS, Grafana, Prometheus.

AI, Architecture & Other: Anthropic LLMs, OpenAPI, TensorFlow/Keras, OpenCV, IPFS, Blockchain, Agile/Scrum

REFERENCES

- Mr. Danesh Kuruppu, Technical Lead | WSO2
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- Mr. Nethandun Wijethilake, Pre-Sales Manager - South Asia | Broadpeak
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